

Japan Autos & Semiconductors

Japan Autos & Semiconductors: Who could be Denso's next semiconductor acquisition target?



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Since Denso's acquisition proposal for Rohm was withdrawn at the end of April, a number of investors continue to show interest and raise questions about Denso's next steps. We believe M&A remains the most realistic and effective option to achieve the growth targets outlined in its mid-term plan. In this report, we examine Denso's semiconductor business strategy, and assess the strategic options potentially available to the company.

Denso to expand into industrial and consumer semis: Denso's semi business is entirely captive and centered on Power & Analog, including Si-IGBT, SiC, and battery management ASICs. Semiconductor revenue was equivalent to ~JPY 420 bn in FY3/22, with Discrete accounting for ~74% of revenue and ASICs for ~16%. The mid-term plan highlights expansion of its semi business into industrial/consumer electronics. As Denso's minority investments and partnerships with power semi majors have focused on strengthening its auto business, we believe a target with industrial and consumer exposure is key.

Denso's target segments look attractive: While overall semi market growth is driven by AI-related memory and ASIC/ASSP, Denso's target segments still are appealing. Across 9 devices x 7 applications combinations, Denso's target 6 segments (Discrete/Analog x Automotive/Industrial/Consumer) offer reasonable scale, relatively high and stable growth, and room for consolidation. The combined market size is expected to grow from USD 50 bn (2025) to USD 80 bn (2030), implying a CAGR of 10%.

Who could be the best fit? We view Fuji Electric (NC) as the most feasible option in terms of acquisition cost, device/application portfolio and management fit. It has the lowest market cap among the major power semi candidates at ~JPY 2 tn, with a long relationship in IGBT/SiC development and meaningful industrial exposure. For the Rohm/Toshiba/MELCO consortium (All NC), Denso might be able to take a partial stake, but a full acquisition would likely be operationally difficult, in our view. A hostile acquisition of Rohm would also be difficult given the reputational risk. While Onsemi (NC) has an attractive product portfolio, we think fully acquiring Onsemi, Renesas, Infineon/STM (NC) looks financially difficult.

Implications for Japan Autos: Automakers are internalizing H/W-linked S/W layers where Denso has been strong amid the shift to SDVs making expansion in semiconductors important. Cost reduction of power semi is critical for improving Toyota's xEV profitability, which would potentially make Denso's M&A reasonable. But using Toyota stake proceeds for M&A could disappoint investors expecting shareholder returns, while buybacks would be positive for Toyota. Reiterate **Market-Perform/Outperform** on **Denso/Toyota**.

Implications for Japan/EU Semiconductors: We are constructive on power semi given the rise of AI power ([Link](#)). We think a Fuji Electric transaction would likely have a limited impact on Renesas, as Denso's benefit would be concentrated in power semis. A Rohm/Toshiba structure could be more relevant for Renesas by strengthening Denso's domestic power semi platform and reducing external sourcing. However, Renesas's MCU leadership and broad customer base would likely limit core business risk. The impact on Infineon would likely also be limited. Reiterate **Outperform** on **Renesas** and **Infineon**.

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	7 Jul 2026	Price	TTM	Reported EPS				Reported P/E (x)		
			Closing	Target	Rel. Perf.	Cur	2025A	2026E	2027E	2025A	2026E	2027E
6902.JP (Denso)	M	JPY	1,985.00	2,050.00	(45.3)%	JPY	162.96	169.90	190.05	12.2	11.7	10.4
7203.JP (Toyota)	O	JPY	2,946.00	4,200.00	(28.5)%	JPY	295.25	308.93	367.29	10.0	9.5	8.0
6723.JP (Renesas)	O	JPY	4,744.00	6,300.00	107.4%	JPY	181.61	244.76	298.54	26.1	19.4	15.9
IFX.GR (Infineon)	O	EUR	76.71	102.00	78.1%	EUR	1.38	1.74	2.94	55.4	44.0	26.1
JPL			2,681.91									
EDME			1,617.02									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

6723.JP, IFX.GR estimate is Adjusted EPS; 6723.JP, IFX.GR valuation is Adjusted P/E (x);

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

Japan Autos and Auto Parts:

As the industry shifts toward SDVs, automakers are increasingly seeking control over hardware-linked upstream software layers, where suppliers such as Denso have historically captured value through integrated hardware and embedded software offerings. This shift could reduce the scope for supplier-led system design over time, making Denso's expansion into upstream hardware such as semiconductors, as well as non-auto businesses, strategically important. For Toyota, cost reduction of power semiconductors, particularly SiC, is important for improving its xEV competitiveness, which would make a potential acquisition by Denso's of a power semiconductor company strategically reasonable. However, if Denso were to use proceeds from a potential sale of its Toyota stake to fund semiconductor M&A, this could disappoint Denso's investors expecting shareholder returns. Conversely, if Toyota were to buy back those shares, it would be positive for Toyota. We remain **Market-Perform** on **Denso** (PT = 2,050.00) and **Outperform** on **Toyota** (PT=4,200.00).

Japan and European Semiconductors:

We are constructive on power semi given the rise of AI power ([Infineon/Renesas: CPU renaissance emerges as a second engine for power semi demand, set to outstrip supply](#)). We believe a successful Fuji Electric transaction would likely limit direct impact on Renesas, as the strategic benefit for Denso would be concentrated in power semis rather than in Renesas's core MCU franchise. A Rohm/Toshiba-related structure could be more relevant for Renesas, as it would give Denso a broader domestic power semiconductor platform and potentially reduce external sourcing needs in selected auto discrete and analog areas. However, Renesas remains much stronger in automotive MCUs and has a broader customer base across OEMs and Tier-1s, limiting the risk of meaningful share loss in its core business. For Infineon, the impact would potentially be limited given only 9% sales exposure to Japanese OEMs, although it remains the second-largest automotive semiconductor supplier in Japan by market share. We rate **Renesas Outperform** with PT = **¥6,300.00** and rate **Infineon Outperform** with PT = **€102.00**.

DETAILS

Since Denso's acquisition proposal for Rohm was withdrawn at the end of April, a number of investors continue to show interest and raise questions about Denso's next steps. We believe M&A remains the most realistic and effective option to achieve the growth targets outlined in its mid-term plan. In this report, we examine Denso's semiconductor business strategy, and assess the strategic options potentially available to the company.

OVERVIEW OF DENSO'S SEMICONDUCTOR BUSINESS

Current state of Denso's semiconductor business

Denso's semiconductor business currently spans three categories: MCU & SoC, Power & Analog, and Sensors (Exhibit 1). At present, all semiconductor products are developed primarily for internal use, supporting Denso's auto parts business. In 'Microcomputer and SoC' subsegment, Denso focuses on architecture design and specification development rather than manufacturing. The company works closely with microcomputer and SoC vendors to develop custom chips tailored to its automotive applications, while wafer fabrication is outsourced to foundries such as TSMC and UMC.

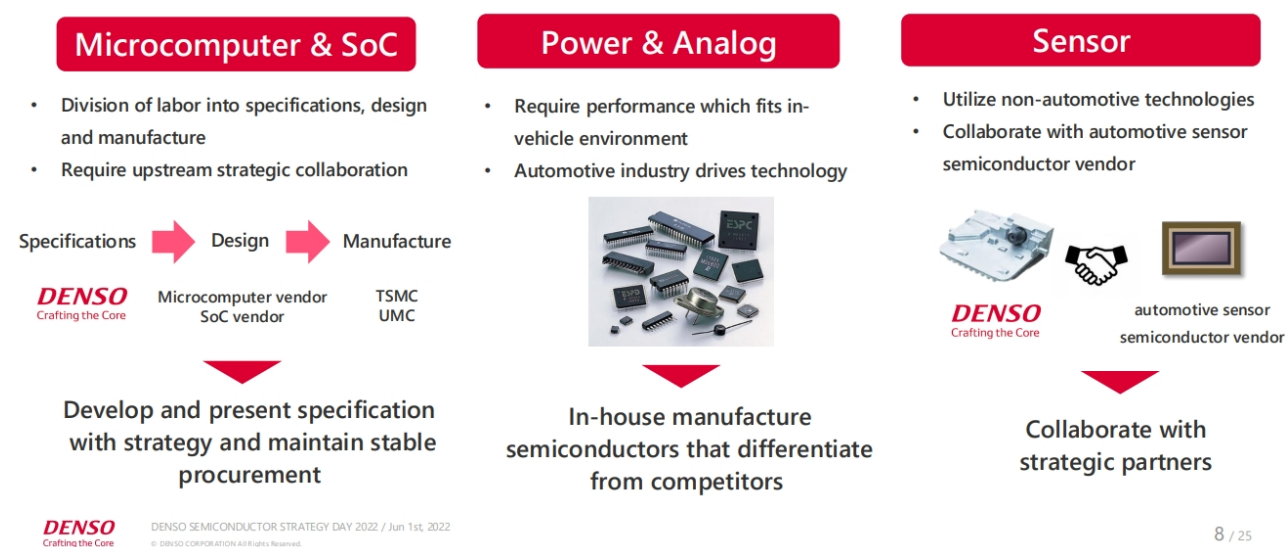
Within 'Sensor' subsegment, Denso develops semiconductor devices that support its ADAS/AD businesses. While the company continues to procure a significant portion of semiconductor content from external suppliers, it develops and manufactures application devices, including image sensors, millimeter-wave radar chips, and LiDAR-related semiconductors.

'Power & Analog' subsegment is the most important subsegment of Denso's semiconductor business, representing the area where the company has internalized the entire value chain from device design to manufacturing. Denso develops and produces key semiconductors for electrification, including Si and SiC power semiconductors used in xEV inverters as well as battery management ASICs. This vertically integrated model enables the company to maintain control over critical technologies while supporting the growing semiconductor content required for automotive electrification.

EXHIBIT 1: Denso's semiconductor business currently spans three categories: MCU & SoC, Power & Analog, and Sensors

Basic Strategy of DENSO's Semiconductor Business

Develop novel and rugged in-vehicle semiconductors while taking full advantage of existing semiconductors, depending on fields.



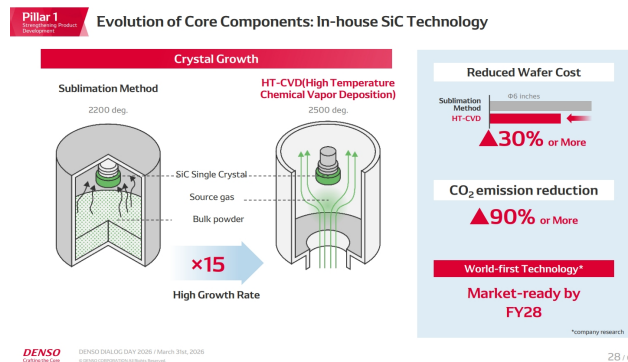
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Source: Denso

Denso's history in power and analog semiconductors dates back several decades. The company began IC research in the 1950s, established its first wafer fabrication facility in 1975, and further expanded its semiconductor capabilities through the acquisition of Toyota's electronic component business in 2020.

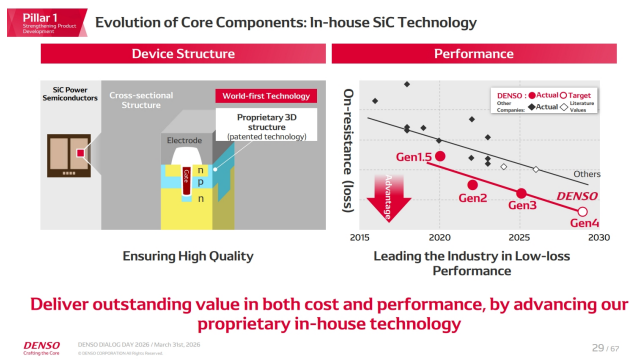
Particularly in power semiconductors, Denso possesses a broad range of proprietary technologies (Exhibit 2, Exhibit 3). In Si power semiconductors, the company jointly developed RC-IGBTs with Fuji Electric, integrating IGBTs and diodes into a single device to achieve smaller size and lower power loss, and has deployed the technology in HEVs. In SiC, Denso has proprietary RAF and gas growth technologies for SiC wafer production, enabling the manufacture of high quality, low cost SiC wafers.

EXHIBIT 2: Denso possesses an in-house SiC crystal growth method called 'Gas Method'



Source: Denso

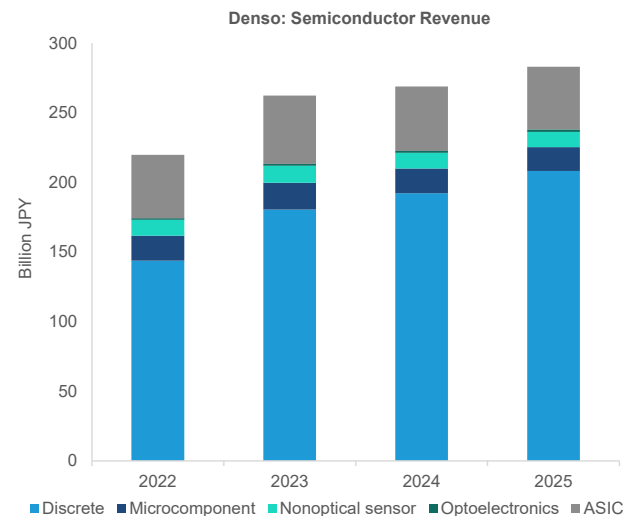
EXHIBIT 3: Denso plans to improve SiC's low-loss performance through several proprietary technologies



Source: Denso

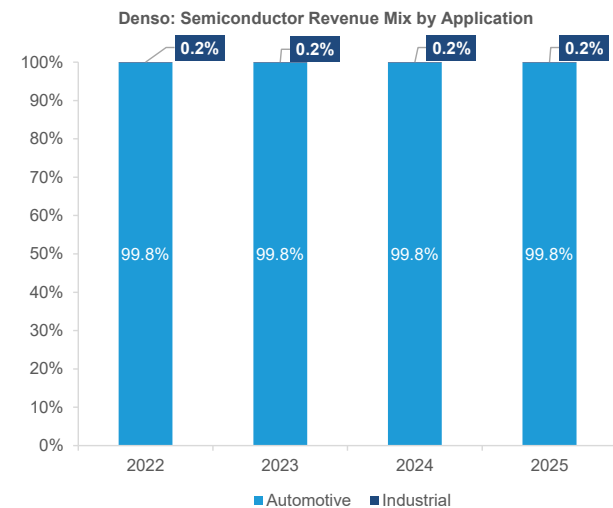
According to Denso's 'Semiconductor Business Briefing' in 2022, the company's semiconductor business generated revenue equivalent to ~JPY 420 bn in FY3/22. Gartner estimates Denso's semiconductor revenue at JPY 208 bn in 2025 (Exhibit 4). By device category, discrete semiconductors, which include power semiconductors, account for 73.6% of revenue, followed by ASICs (Application Specific Integrated Circuits) at 15.9%, Microcomponents, including MCUs, at 6.0%, and Non-optical sensors at 3.9%. As a result, the areas that Denso classifies as 'Power & Analog' represent ~90% of total semiconductor revenue. By application, the business is almost entirely exposed to the automotive market, which accounts for 99.7% of revenue, with only 0.3% attributable to industrial applications (Exhibit 5).

EXHIBIT 4: Gartner estimates Denso's semiconductor revenue at JPY 208 bn in 2025



Source: Gartner, Bernstein analysis

EXHIBIT 5: Denso's semiconductor business is almost entirely exposed to the automotive market



Source: Gartner, Bernstein analysis

Denso's semiconductor business strategy

Denso identified semiconductors as one of its 'Societal Value Expansion Domains' in the mid-term plan announced in Mar-2026 (Exhibit 6). The company aims to expand the application of its semiconductor technologies beyond automotive markets and 'contribute to a wide range of fields, including industrial equipment and consumer electronics'. Given that Denso currently generates no external semiconductor revenue and derives nearly all of its semiconductor sales from automotive applications, we believe that power semiconductor companies with meaningful exposure to industrial and consumer end markets could be attractive strategic targets for the company.

EXHIBIT 6: Denso identified semiconductors as one of its 'Societal Value Expansion Domains' in the mid-term plan announced in Mar-2026



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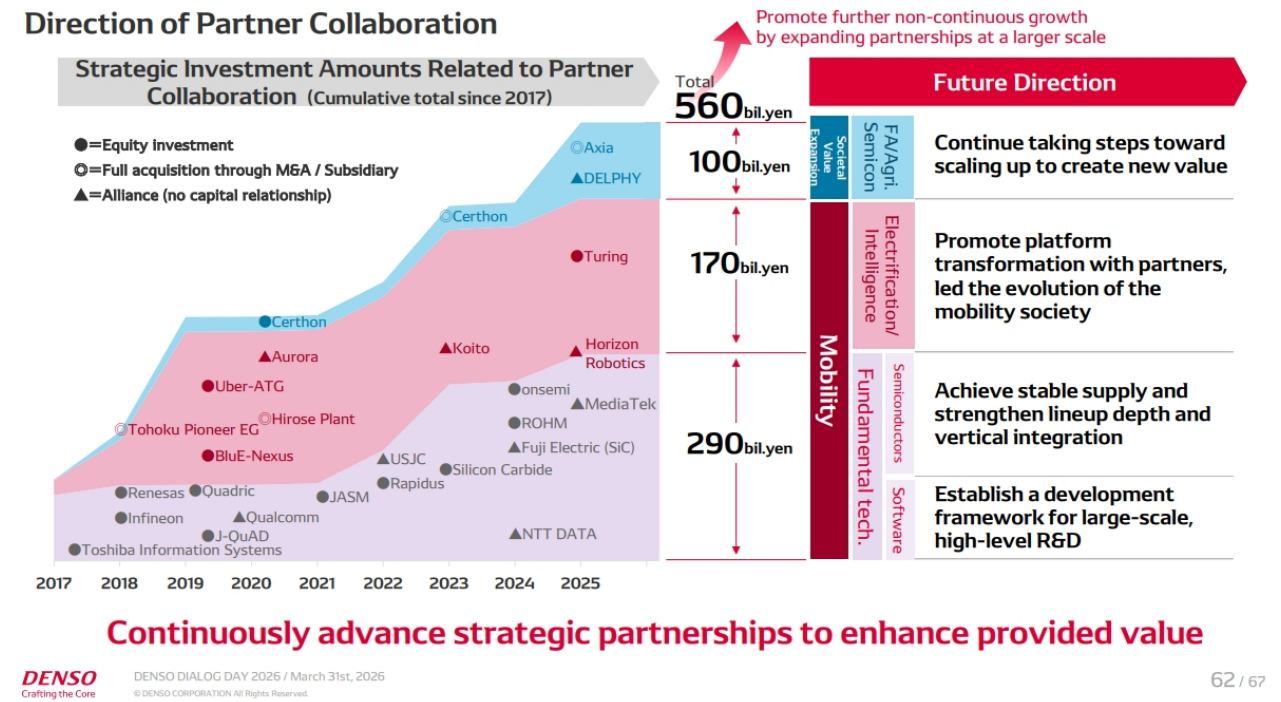
Source: Denso

Denso has historically pursued minority equity investments to strengthen semiconductor supply security and strategic partnerships (Exhibit 7). Investments have included Infineon and Renesas in 2018, JASM (private, TSMC's Japanese manufacturing subsidiary), in 2022, Coherent's SiC wafer business in 2023, onsemi (not covered) in 2024, and Rohm (not covered) in 2025.

Beyond equity investments, Denso has also established a broad range of strategic partnerships. These include Qualcomm for next-generation cockpit systems in 2020, USJC for Si power semiconductor manufacturing at UMC's Japanese fab in 2022, Fuji Electric for joint SiC production in 2024, MediaTek (covered by Mark Li) for SoC development in 2025, and Horizon Robotics (covered by Qingyuan Lin) for ADAS and autonomous driving systems targeting the Chinese market in 2025.

While Denso has built deep relationships with a diverse set of semiconductor companies through minority investments and partnerships, these initiatives have largely been aimed at strengthening its automotive business. In the context of semiconductor M&A, however, acquisition targets that can deliver meaningful inorganic growth beyond the automotive market are likely to be of greater strategic importance, in our view.

EXHIBIT 7: Denso has historically pursued minority equity investments to strengthen semiconductor supply security and strategic partnerships



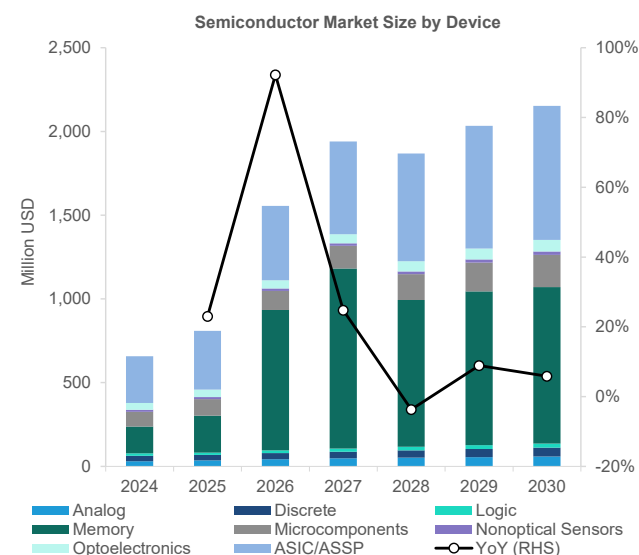
Source: Denso

MARKET ATTRACTIVENESS OF DENSO'S TARGET SEMICONDUCTOR SEGMENTS

According to Gartner, the global semiconductor market is expected to grow from USD 809 bn in 2026 to USD 2,153 bn in 2030, implying a CAGR of 21.6% (Exhibit 8). Growth is expected to be driven primarily by memory and ASIC/ASSP markets, which benefit from rising AI-related demand, including discrete GPUs and AI accelerators (Exhibit 9). Combined, these categories are expected to account for more than 80% of the semiconductor market throughout the period from 2026 onward.

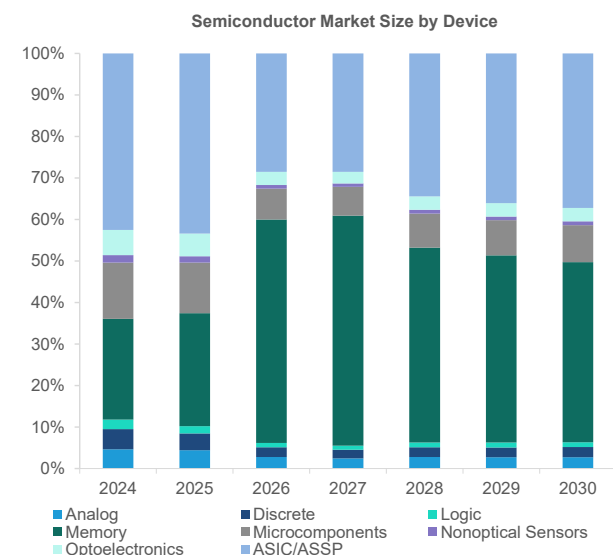
By comparison, the discrete and analog markets, which represent Denso's primary areas of focus, are expected to account for only around 5% of the overall semiconductor market. Nevertheless, the combined market size is projected to increase steadily from USD 79 bn in 2026 to USD 111 bn in 2030, making them attractive markets characterized by stable growth rather than rapid expansion.

EXHIBIT 8: The global semiconductor market is expected to grow from USD 0.8 tn in 2026 to USD 2.2 tn in 2030, implying a CAGR of 21.6%



Source: Gartner, Bernstein analysis

EXHIBIT 9: Growth is expected to be driven primarily by memory and ASIC/ASSP markets, which benefit from rising AI-related demand



Source: Gartner, Bernstein analysis

By application, the 'Data processing (compute)' segment, which includes chips for AI servers, is expected to remain the primary driver of semiconductor market growth, with its share of the global semiconductor market projected to exceed 50% by 2030 (Exhibit 10, Exhibit 11). In contrast, the 'Automotive, Industrial, and Consumer markets, which represent Denso's primary target end markets, are expected to account for ~20% of the semiconductor market throughout 2030. Nevertheless, the combined market is projected to grow steadily from USD 327 bn in 2026 to USD 411 bn in 2030, providing a stable growth opportunity.

EXHIBIT 10: The 'Data processing (compute)' segment, which includes chips for AI servers, is expected to remain the primary driver of market growth

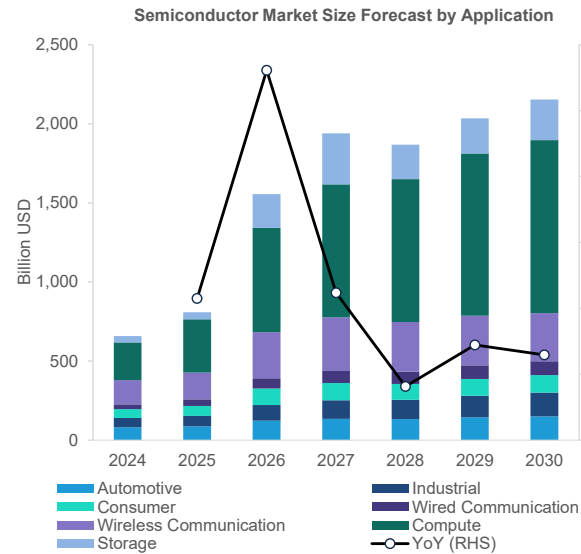
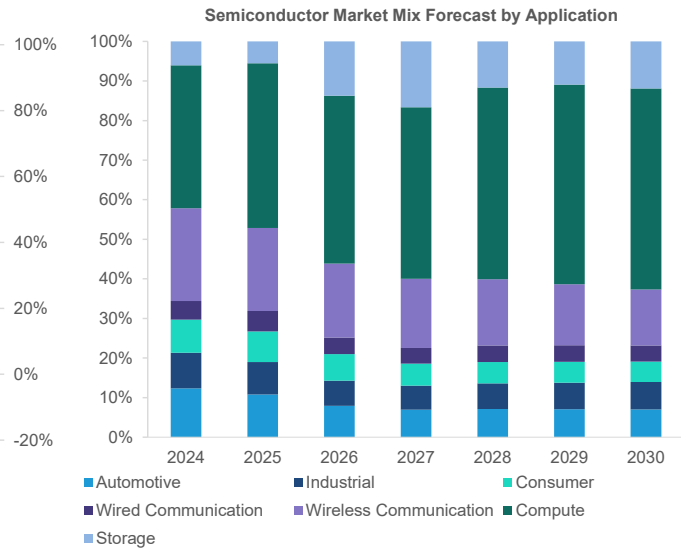


EXHIBIT 11: The share of the 'Data processing (Compute)' application category out of total semiconductor market is projected to exceed 50% by 2030

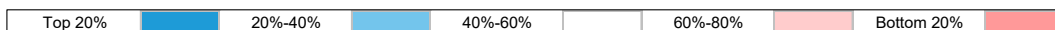


We assess the attractiveness of the 63 semiconductor segments formed by the combination of 9 device categories and 7 application categories. Our analysis focuses on four metrics: market size, CAGR (2025–2030), volatility (measured as the standard deviation of annual growth rates between 2025 and 2030), and market concentration (measured by the combined market share of the top three vendors in each segment).

In terms of market size, Discrete × Automotive, Discrete × Industrial, and Analog × Industrial rank in the top 20-40% of the 63 device-application combinations, suggesting that they are relatively large semiconductor markets (Exhibit 12).

EXHIBIT 12: Discrete × Automotive, Discrete × Industrial, and Analog × Industrial rank in the top 20-40% of the 63 device-application combinations,

Market Size (2025, USD MM)	Automotive	Industrial	Consumer	Wired Communication	Wireless Communication	Data processing (Compute)	Data processing (Storage)	Total by Application
Analog	5,727	12,352	4,198	2,581	4,646	5,783	297	35,584
Discrete	13,376	10,206	3,972	552	1,967	2,464	154	32,691
Logic	1,617	3,560	2,516	792	3,881	2,046	125	14,537
Memory	12,128	7,907	17,861	4,278	53,864	88,679	35,338	220,055
Microcomponent	11,879	6,569	3,043	4,122	1,348	70,653	730	98,344
Nonoptical sensor	5,058	1,905	1,803	43	3,141	718	27	12,695
Optoelectronics	7,484	9,648	2,892	3,976	17,860	2,275	120	44,255
ASIC	2,732	2,805	7,515	6,798	16,774	28,491	2,578	67,693
ASSP	27,305	11,183	18,910	18,732	66,002	135,622	5,422	283,176
Total by Device	87,306	66,135	62,710	41,875	169,483	336,731	44,792	809,031



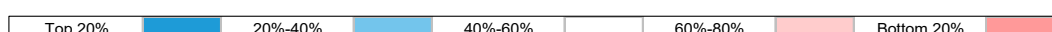
Source: Gartner, Bernstein analysis

In terms of growth, Industrial markets stand out as particularly attractive. Industrial × Analog and Industrial × Discrete are expected to deliver CAGRs of 13.2% and 13.0%, respectively, making them among the fastest-growing segments across the 63 device-application combinations (Exhibit 13). At the same time, these markets exhibit relatively low volatility in annual growth rates. In particular, Discrete × Industrial and Discrete × Consumer show volatility of just 3.1% and 1.2%, respectively, suggesting the potential for stable long-term growth (Exhibit 14). This contrasts with memory markets, which offer stronger

growth potential but are typically subject to significantly higher volatility due to semiconductor cycles.

EXHIBIT 13: Industrial × Analog and Industrial × Discrete are expected to deliver CAGRs of 13.2% and 13.0%, respectively

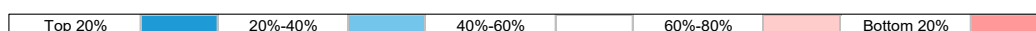
CAGR (2025-2030)	Automotive	Industrial	Consumer	Wired Communication	Wireless Communication	Data processing (Compute)	Data processing (Storage)	Total by Application
Analog	3.2%	13.2%	1.0%	16.1%	5.0%	17.2%	9.0%	10.5%
Discrete	10.8%	13.0%	4.5%	4.6%	4.0%	6.7%	-0.6%	10.0%
Logic	10.8%	15.6%	11.2%	9.3%	8.0%	14.9%	5.9%	11.9%
Memory	24.3%	34.1%	19.5%	21.5%	23.7%	35.9%	46.9%	33.5%
Microcomponent	5.5%	20.2%	16.2%	6.7%	10.3%	15.5%	-2.3%	14.4%
Nonoptical sensor	6.5%	22.4%	11.0%	3.3%	2.6%	4.2%	1.1%	9.2%
Optoelectronics	10.7%	10.6%	8.7%	20.2%	4.2%	3.0%	78.5%	9.5%
ASIC/ASSP	9.9%	15.2%	8.5%	16.1%	5.2%	25.6%	6.0%	17.9%
Total by Device	11.5%	17.7%	12.0%	16.1%	12.4%	26.6%	41.7%	21.6%



Source: Gartner, Bernstein analysis

EXHIBIT 14: These markets exhibit relatively low volatility in annual growth rates, where Discrete × Industrial and Discrete × Consumer show volatility of just 3.1% and 1.2%, respectively

Volatility (2025-2030)	Automotive	Industrial	Consumer	Wired Communication	Wireless Communication	Data processing (Compute)	Data processing (Storage)	Total by Application
Analog	4.7%	7.4%	2.8%	14.4%	2.5%	11.3%	4.5%	5.8%
Discrete	6.1%	3.1%	1.2%	0.9%	4.1%	5.8%	2.4%	3.3%
Logic	8.6%	6.1%	6.2%	15.6%	7.6%	6.0%	5.1%	6.4%
Memory	74.2%	98.4%	86.3%	95.7%	85.5%	92.4%	175.2%	102.2%
Microcomponent	4.4%	7.6%	5.5%	10.9%	11.5%	3.1%	3.1%	2.6%
Nonoptical sensor	2.7%	6.7%	3.2%	4.6%	3.0%	3.5%	6.3%	2.1%
Optoelectronics	6.5%	1.8%	2.3%	16.9%	5.2%	2.8%	91.4%	2.0%
ASIC/ASSP	2.9%	2.7%	5.9%	12.6%	3.6%	14.4%	5.0%	6.5%
Total by Device	13.7%	14.9%	24.3%	19.0%	26.8%	31.0%	138.5%	31.6%



Source: Gartner, Bernstein analysis

In terms of market concentration, Discrete × Automotive, Discrete × Industrial, and Discrete × Consumer remain relatively fragmented markets, suggesting significant potential for industry consolidation (Exhibit 15). This also stands in contrast to the memory market, where a small number of leading players already dominate the industry.

Taken together, these characteristics suggest that Denso's focus on Analog and Discrete semiconductors across automotive, industrial, and consumer applications is strategically sound. These markets offer a combination of reasonable scale, relatively high and stable growth, and substantial consolidation opportunities, making them attractive targets for long-term expansion.

EXHIBIT 15: Discrete × Automotive, Discrete × Industrial, and Discrete × Consumer remain relatively fragmented markets, suggesting significant potential for industry consolidation

Concentration (2025)	Automotive	Industrial	Consumer	Wired Communication	Wireless Communication	Data processing (Compute)	Data processing (Storage)	Total by Application
Analog	58.8%	67.2%	25.2%	73.7%	37.4%	58.9%	67.6%	53.9%
Discrete	40.4%	38.8%	28.2%	41.3%	41.7%	38.8%	54.9%	30.8%
Logic	49.7%	59.2%	44.4%	80.4%	58.0%	34.3%	68.8%	39.0%
Memory	79.5%	70.4%	72.7%	73.3%	79.3%	92.6%	65.0%	80.5%
Microcomponent	75.0%	47.4%	30.1%	76.8%	76.8%	88.7%	96.4%	69.3%
Nonoptical sensor	46.2%	23.8%	58.0%	70.4%	39.7%	33.0%	84.7%	31.5%
Optoelectronics	41.9%	28.0%	21.4%	66.6%	74.5%	48.4%	47.2%	42.5%
ASIC	51.7%	27.3%	62.8%	62.2%	88.6%	79.8%	84.4%	63.4%
ASSP	36.9%	21.0%	34.3%	77.1%	67.1%	87.6%	49.4%	58.1%
Total by Device	26.0%	20.8%	23.8%	57.7%	45.9%	61.3%	46.0%	32.7%

Top 20%		20%-40%		40%-60%		60%-80%		Bottom 20%
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Source: Gartner, Bernstein analysis

EVALUATING DENSO'S POTENTIAL SEMICONDUCTOR M&A OPTIONS

Based on the analysis above, we assess the semiconductor M&A options potentially available to Denso. Given that Rohm was the company's most recent acquisition target, we focus our analysis on major power semiconductor vendors. We exclude analog semiconductor companies from consideration, as the leading players in the segment, such as Texas Instruments and Analog Devices (both covered by Stacy Rasgon), are substantially larger than Denso's realistic acquisition universe. In addition, Denso's semiconductor business remains primarily focused on power semiconductors, making power semiconductor vendors the most relevant strategic targets, in our view. Among Japanese power semiconductor companies, we focus on Fuji Electric, Rohm, Mitsubishi Electric (MELCO, not covered), Toshiba (private), and Renesas, while onsemi (not covered), Infineon, and STMicroelectronics (not covered) represent potential US and European candidates (Exhibit 16).

Fuji Electric appears to be Denso's closest semiconductor partner. In Si power semiconductors, the two companies jointly developed RC-IGBTs, and we believe the majority of Fuji Electric's automotive semiconductor business is currently tied to Denso. The relationship has deepened further through their planned joint production of SiC power semiconductors, which is scheduled to begin in 2H CY2026. Taken together, the long-standing technological collaboration, manufacturing partnership, and existing business ties suggest a particularly high degree of strategic and operational fit between the two companies.

Rohm became a strategic partner of Denso in 2025 through a collaboration focused on analog ICs and automotive semiconductors, accompanied by Denso's acquisition of ~5% stake in the company. In March 2026, Denso submitted a proposal to take Rohm private, which was subsequently rejected by Rohm's management. Rohm has also been working with **Toshiba** since 2023 to explore collaboration in power semiconductors, although progress has been limited. Following Denso's approach, Rohm expanded the discussions to include **Mitsubishi Electric** and has been exploring a potential business and management integration among the three companies ([link](#)). However, little visible progress has been made over the past two months. For Denso, both Mitsubishi Electric and Toshiba currently maintain conventional business relationships as suppliers of power semiconductors and other components.

Renesas is one of the world's leading automotive MCU suppliers and has long maintained close ties with Toyota and Denso, reflecting its historical origins. Denso increased its ownership stake in Renesas from ~0.5% to 5% in 2018, before gradually reducing it to around 0.1% by 2025. Through a series of acquisitions, Renesas has expanded its presence in analog semiconductors while also strengthening its power semiconductor portfolio. However, the company previously abandoned plans to establish in-house SiC production capabilities.

Among US and European power semiconductor companies, Denso maintains equity relationships with several players. In 2024, Denso acquired a stake in **Onsemi** and currently owns ~0.05% of the company. The investment was made in connection with their collaboration in ADAS/AD semiconductors. Similarly, Denso acquired a stake in **Infineon** in 2018 to strengthen cooperation in automotive semiconductors and currently holds ~0.2% of the company. Following the investment, the two companies have reportedly deepened their business relationship through regular management-level discussions and other collaborative activities. **STMicroelectronics** is the only one of the three major power semiconductor vendors in which Denso does not hold an equity stake. As such, the relationship appears to be limited to a conventional customer-supplier partnership.

EXHIBIT 16: Among Japanese power semiconductor companies, we focus on Fuji Electric, Rohm, Mitsubishi Electric, Toshiba, and Renesas, while onsemi, Infineon, and STM represent potential US and European candidates

		Financials (JPY Bn)	Overview / Relationship with Denso
----> Capital relationship —> Business relationship	 Fuji Electric	Mkt Cap: 2,097 Revenue: 1,228 OP (OPM): 137 (11%)	• Closest power semi partner through the joint development of Si RC-IGBTs and planned SiC production in Japan (from Sep-26) • Most of Fuji Electric's automotive semiconductor revenue (~50% of semiconductor sales) is believed to be derived from Denso
	Integration ROHM	Mkt Cap: 2,222 Revenue: 481 OP (OPM): 11 (2%)	• Entered a strategic partnership in analog ICs and automotive semiconductors in 2025, with Denso acquiring a stake of just under 5%, while Denso proposed a take-private offer in 2026 (failed) • Currently in discussions with Mitsubishi Electric and Toshiba on a potential business and management integration of their power device operations
	 TOSHIBA	Mkt Cap: 2,000*1 Revenue: 3,362 OP (OPM): 111 (3%)	• Active in Si power semiconductor and is one of Denso's key power semiconductor suppliers • JPY 300 bn funding from Rohm for Toshiba's privatization in 2023 • Currently in discussions with Mitsubishi Electric and Toshiba on a potential business and management integration of their power device operations
	 MITSUBISHI ELECTRIC	Mkt Cap: 12,751 Revenue: 5,895 OP (OPM): 433 (7%)	• Active in Si and SiC power semiconductors and is one of Denso's key power semiconductor suppliers • Currently in discussions with Rohm and Toshiba on a potential business and management integration of their power device operations
	 RENESAS	Mkt Cap: 9,583 Revenue: 1,393 OP (OPM): 270 (19%)	• Active in Si and GaN power semiconductors and is one of Denso's key semiconductor suppliers • Scrapped its SiC production plans following Wolfspeed's bankruptcy • Denso increased its stake in Renesas from 0.1% to 5% in 2018, but has since reduced its ownership back to 0.1%
	 Infineon	Mkt Cap: 19,943 Revenue: 2,417 OP (OPM): 250 (10%)	• Active in Si, SiC and GaN power semiconductors and is one of Denso's key semiconductor suppliers • Denso acquired a stake in 2018 to strengthen the relationship and currently holds a 0.2% stake • Business ties with Denso have deepened since the investment through regular management interactions
	 onsemi	Mkt Cap: 6,028 Revenue: 897 OP (OPM): 234 (26%)	• Active in Si, SiC and GaN power semiconductors and is one of Denso's key semiconductor suppliers • Denso acquired a stake in 2024 to strengthen the partnership in ADAS/AD semiconductors and currently holds a 0.05% stake
	 ST	Mkt Cap: 11,031 Revenue: 1,865 OP (OPM): 137 (5%)	• Active in Si, SiC and GaN power semiconductors and presumably is one of Denso's key semiconductor suppliers • No capital relationship with Denso

Note 1: Market cap at the time of privatization in 2023

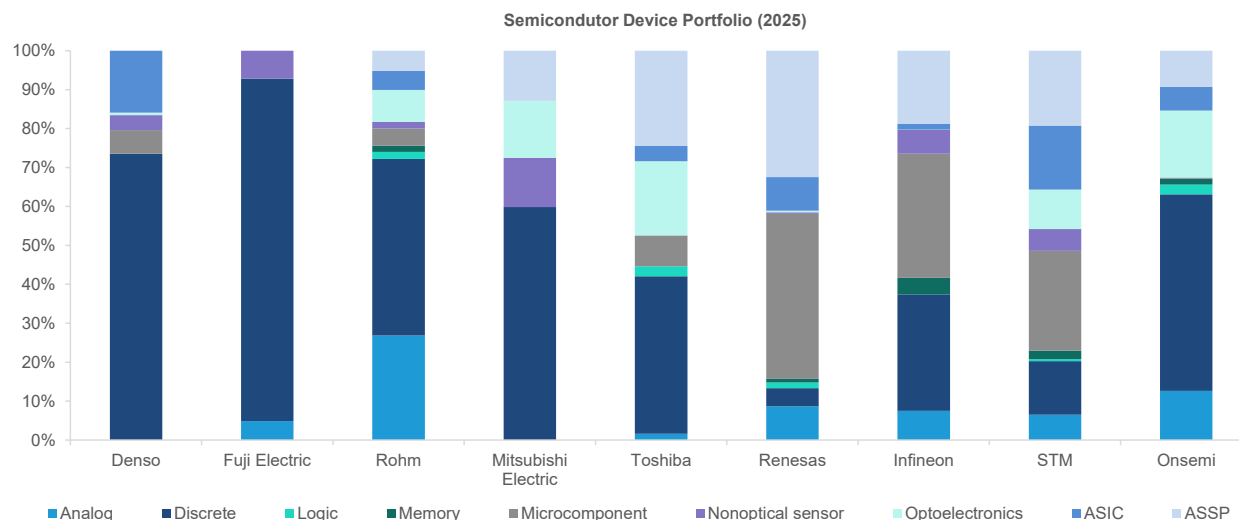
Source: Bloomberg, Company reports, Bernstein analysis

To assess strategic fit, we first examine each company's semiconductor portfolio by device category (Exhibit 17). Fuji Electric is effectively a pure-play power semiconductor company, with Discrete devices accounting for 88% of semiconductor revenue. In contrast, Rohm maintains a well-balanced portfolio centered on Analog and Discrete semiconductors while also participating in a broad range of other semiconductor categories, highlighting why the company has attracted Denso's interest.

Mitsubishi Electric derives ~60% of its semiconductor revenue from discrete devices, while also maintaining meaningful exposure to communications-related semiconductors, including Nonoptical sensors and Optoelectronics. Toshiba's portfolio is similarly diversified, with Discrete devices accounting for roughly 40% of revenue and Optoelectronics representing ~19%, resulting in a balanced mix across multiple semiconductor categories.

Renesas stands apart from the other power semiconductor players. Microcomponents account for 42% of its semiconductor revenue, followed by ASSPs at 32%, making Analog and Discrete products relatively minor parts of its overall portfolio.

Infineon and STMicroelectronics also maintain diversified portfolios, although Analog is the largest category for both companies, accounting for roughly 30% of revenue. While both companies have meaningful exposure to power semiconductors, their product mix extends well beyond the area of Denso's core focus. Onsemi presents a different profile. Discrete devices account for ~51% of revenue, followed by Optoelectronics at 17% and Analog at 13%. This portfolio is more comparable to those of Japanese power semiconductor vendors, particularly Toshiba. As a result, onsemi's product mix appears strategically attractive from Denso's perspective.

EXHIBIT 17: **Semiconductor device portfolio by candidate as of 2025**

Source: Gartner, Bernstein analysis

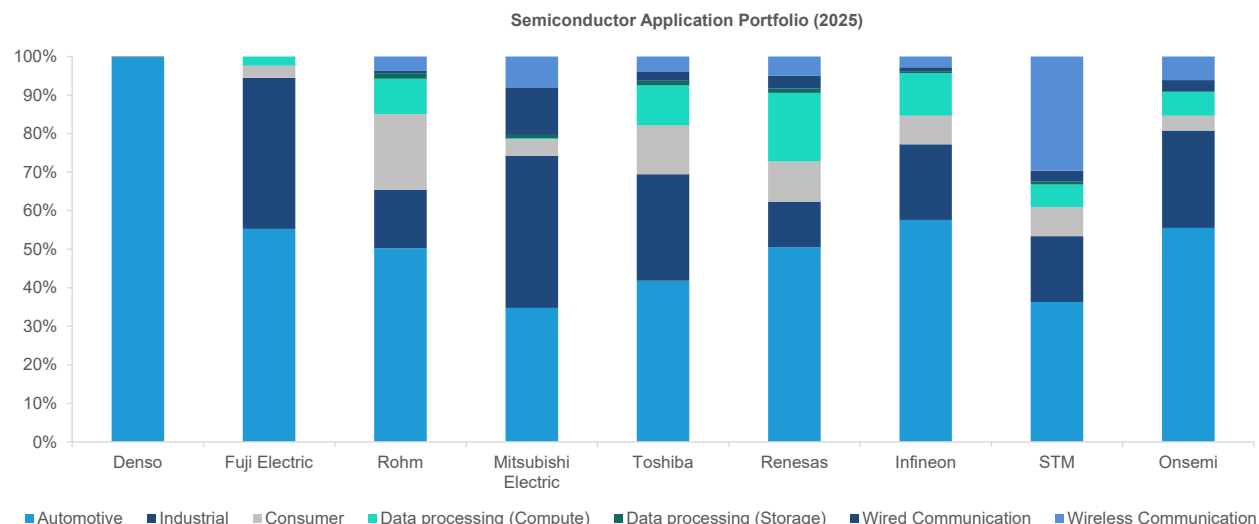
By application, automotive remains the largest end market across all of the companies under review, although exposure outside automotive varies meaningfully by player (Exhibit 18). Fuji Electric derives ~39% of semiconductor revenue from Industrial and only ~3% from Consumer, resulting in a portfolio that is effectively split between automotive and industrial end markets.

Rohm has relatively greater exposure to Consumer, with Consumer accounting for ~20% of revenue and Industrial representing ~15%. Mitsubishi Electric benefits from internal demand from its own industrial businesses, resulting in a well-balanced application mix of roughly 35% Automotive, 40% Industrial, 4% Consumer, and 12% Wired Communication. Toshiba exhibits a similar profile, supported by internal demand from group businesses. Automotive accounts for ~42% of revenue, followed by Industrial at 28%, Consumer at 13%, and Data Processing (Compute) at around 10%.

Renesas generates ~51% of revenue from Automotive, but its second-largest market is Data Processing (Compute) at 18%, followed by Industrial at 12% and Consumer at 10%. This suggests a growing exposure to compute-related markets that are benefiting from AI-driven demand.

Infineon and onsemi have relatively similar application portfolios. Infineon derives ~58% of revenue from Automotive, 20% from Industrial, and 7% from Consumer applications, while Onsemi generates 56%, 25%, and 4%, respectively. Given Denso's objective of expanding into industrial and consumer applications beyond automotive, both companies offer attractive end-market exposure.

STMicroelectronics stands out from the peer group. While Automotive remains its largest application at ~36% of revenue, Wireless Communication accounts for ~30%, making it the only company among the candidates with such significant exposure to smartphone and mobile communication markets.

EXHIBIT 18: **Semiconductor application portfolio by candidate as of 2025**

Source: Gartner, Bernstein analysis

Finally, we assess Denso's potential strategic options. Our evaluation is based on four criteria: acquisition cost, device portfolio, application portfolio, and strategic and management fit with Denso (Exhibit 19).

We believe the most likely scenario could be a full acquisition of Fuji Electric. Among the potential targets, Fuji Electric has the lowest market cap at ~JPY 2.0 tn. Even assuming a 40% acquisition premium, the implied transaction value of JPY 2.8 tn remains within Denso's estimated funding capacity of JPY 4.0 tn. In addition, the two companies have built a close business relationship over many years. Fuji Electric's focus on power semiconductors, supplemented by selective analog products and meaningful exposure to industrial applications, aligns well with Denso's strategic objective of expanding beyond the automotive market. As a result, Fuji Electric appears to be the most feasible and strategically attractive acquisition candidate.

The second scenario could be for Denso to participate in the ongoing discussions among Rohm, Toshiba, and Mitsubishi Electric regarding the integration of their power semiconductor businesses. Based on a rough calculation using each company's market cap and the proportion of revenue attributable to semiconductors, we estimate the combined entity would be worth ~JPY 3.1 tn and JPY 4.3 tn with a 40% premium. While larger than Fuji Electric, such a transaction would still appear financially feasible for Denso, even under a full acquisition scenario. As such, a partial stake acquisition would appear to be the more realistic structure. From a strategic perspective, the combined portfolio would offer a well-balanced mix across Automotive, Industrial, and other end markets, as well as a broad device portfolio. However, we see meaningful execution hurdles. Rohm initially expanded the discussions with Toshiba and Mitsubishi Electric as a response to Denso's acquisition approach, and we believe management opposition to Denso's participation is likely to remain significant. Even if Denso were able to join the discussions, the process would expand into a four-party negotiation, potentially further complicating talks that have shown little visible progress over the past two months.

The next option could be a majority acquisition of Onsemi. Like Rohm, Onsemi combines a strong power semiconductor business with a diversified device and application portfolio, making it an attractive strategic fit for Denso. However, Onsemi's market cap is ~JPY 6.0 tn. Assuming a 40% acquisition premium, a full acquisition would require ~JPY 8.4 tn, making such a transaction unrealistic for Denso. As a result, a full acquisition would likely be unrealistic, leaving a majority stake acquisition as the most feasible structure. Even then, the challenges associated with managing a large US-listed semiconductor company could represent a meaningful hurdle for Denso.

We view a hostile acquisition of Rohm, as well as acquisitions of Renesas, Infineon/STM, as relatively unlikely scenarios. A hostile acquisition of Rohm would likely create meaningful challenges in post-merger integration and management, while also exposing Denso and the broader Toyota group to reputational risk. Renesas appears even less likely given its large market cap, which effectively rules out a full acquisition. In addition, Renesas remains primarily focused on MCUs rather than power semiconductors, and Denso has gradually reduced its ownership stake over the past eight years, suggesting limited strategic and managerial alignment. Similarly, the size of Infineon and STM makes a full acquisition unrealistic. While minority investments

could be conceivable, such investments would be unlikely to materially advance Denso's mid-term strategy.

EXHIBIT 19: We view Fuji Electric as the most feasible option from the perspectives of acquisition cost, device/application portfolio and management fit

#	Acquisition target	Overall Assessment	Acquisition Cost *1	Device Portfolio	Application Portfolio	Management fit	Comment
1	 Fuji Electric	A	A JPY 2.8 tn	A	B	A	- Most affordable target among the candidates (JPY 2 tn market cap), with strong strategic fit driven by long standing collaboration in power device development and manufacturing - High non-semi exposure (~80% of sales), with 'Industry' segment that could also support Denso's FA business expansion strategy
2	  TOSHIBA	B	B JPY 4.3 tn	A	A	C	- Denso may be able to take a partial stake, but a full acquisition would likely be operationally difficult, while a combined power semiconductor consortium would offer an attractive - Denso's participation could further complicate ongoing discussions
3		B	B JPY 8.4 tn	A	A	B	- Onsemi offers a reasonable strategic fit given its power semiconductor portfolio and existing capital relationship with Denso - Limited funding capacity suggests a full acquisition would be challenging, making a majority stake acquisition more likely scenario
4	 Hostile takeover	C	A JPY 2.8 tn	A	A	N/A	While Rohm would be the most attractive target from multiple perspectives, Denso's renewed acquisition attempt would likely face significant management resistance and could damage Denso's reputation
5		C	C JPY 13.4 tn	B	B	C	- Acquisition would be financially demanding, with limited strategic fit given Renesas's focus on analog, MCU and ASIC/ASSP products rather than power semiconductors - Denso has also gradually reduced its ownership stake over time, suggesting limited appetite for deeper integration with Renesas
6	 	C	C >JPY 15 tn	B	B	C	- Infineon and STM are too large for Denso to fully acquire - A minority investment is the most likely outcome, but it would be unlikely to materially support Denso's mid-term plan targets

Note 1: Acquisition cost is calculated as the current market cap + a 40% premium. For the Rohm/Toshiba/Mitsubishi Electric consortium, acquisition cost is estimated as the sum of (i) Rohm's current market cap, (ii) Mitsubishi Electric's market cap multiplied by the semiconductor business revenue contribution, and (iii) Toshiba's implied equity value at the time of its 2023 privatization multiplied by the semiconductor business revenue contribution and (iv) a 40% premium
Source: Bloomberg, Company reports, Bernstein analysis

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VALUATION METHODOLOGY**Denso Corporation**

We set price target of JPY 2,050 using PBR (P/B multiple) of 0.99x against our FY3/28 BPS estimate of JPY 2,112.

Toyota Motor Corporation

We set a price target of JPY 4,200 using reset PBR (P/B multiple) of 1.26x against our FY3/28 BPS estimate of JPY 3,357.

Renesas Electronics Corp

We set a ¥6,300 target price using a P/E multiple of 20x against our forward Q5-Q8 EPS estimate of ¥316.

Infineon Technologies AG

For Infineon, we apply a 30x multiple to our Q5-Q8 EPS estimate of €3.4 to get a PT of €102.

RISKS**Denso Corporation**

Key upside risks to our Denso price target would be 1) exemption on US import parts, 2) aggressive downsizing of ICE-related business, 3) significant improvement in shareholder return policies. Key downside risks would be 1) delay of Toyota's domestic production recovery, 2) weaker than expected vehicle demand, 3) intensified competition in xEV/ADAS systems and components.

Toyota Motor Corporation

Key downside risks would be 1) increase in US import tariffs, 2) slowdown in HEV demand, 3) delay of domestic production recovery.

Renesas Electronics Corp

There are a number of downside risks to our Renesas target price: (1) Prolonged delay in the Analog market recovery (2) Weaker-than-expected market share gain in both Automotive MCU and AI power markets (3) Exchange rate fluctuation with upside / downside risk to overall Japan market.

Infineon Technologies AG

Downside risks to our price target for Infineon include slower automotive demand or EV penetration, push-out of anticipated recovery, price-erosion and oversupply of SiC solutions, lower semiconductor demand broadly as well as valuation multiple compression.

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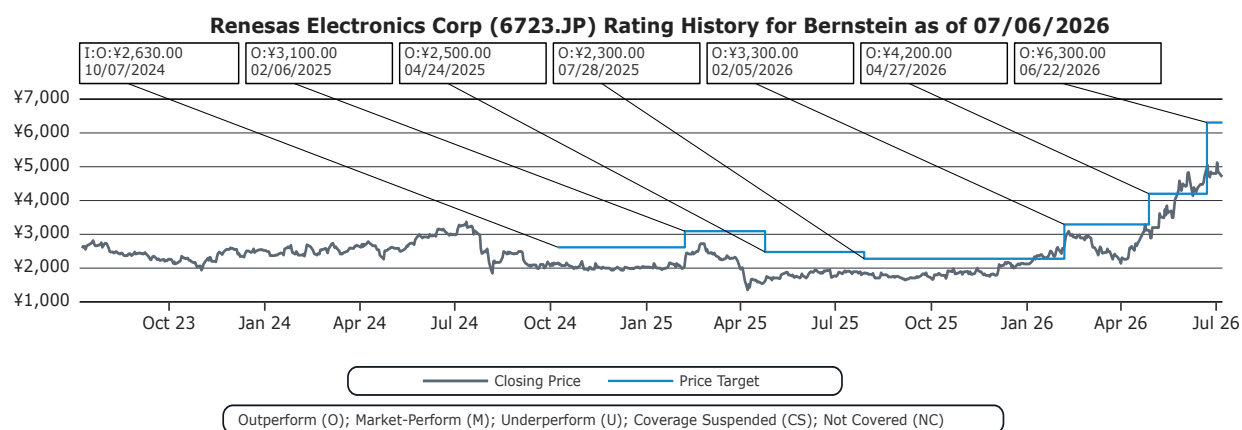
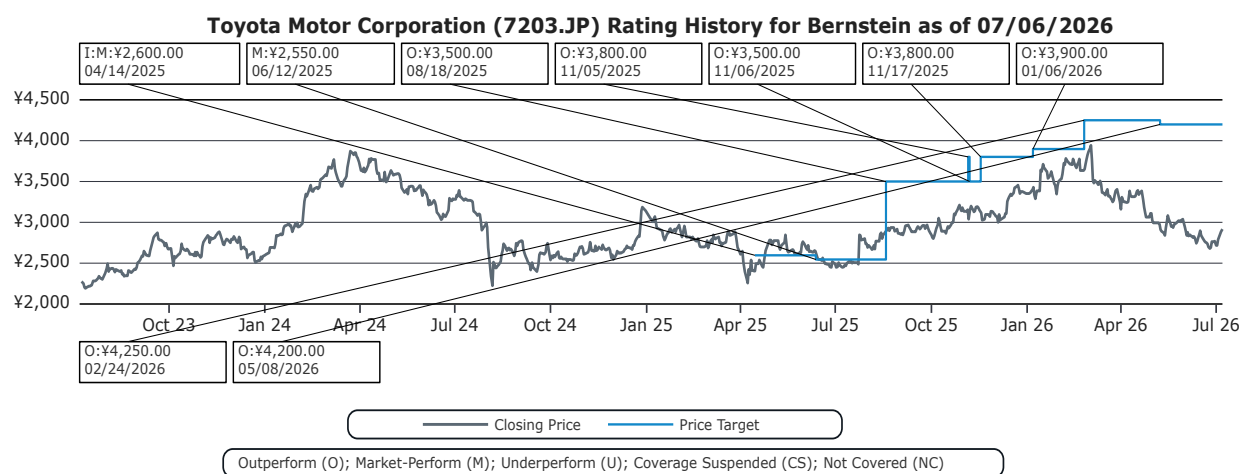
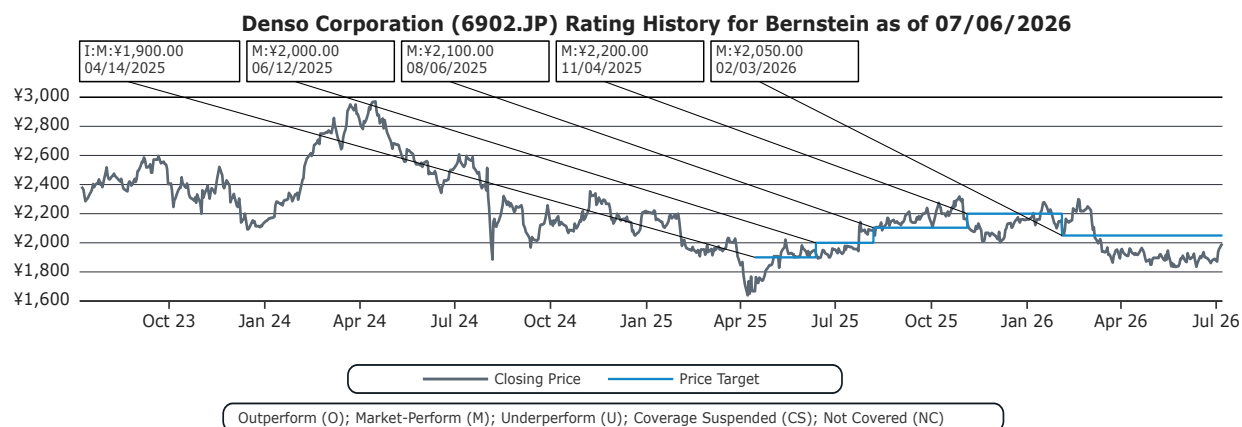
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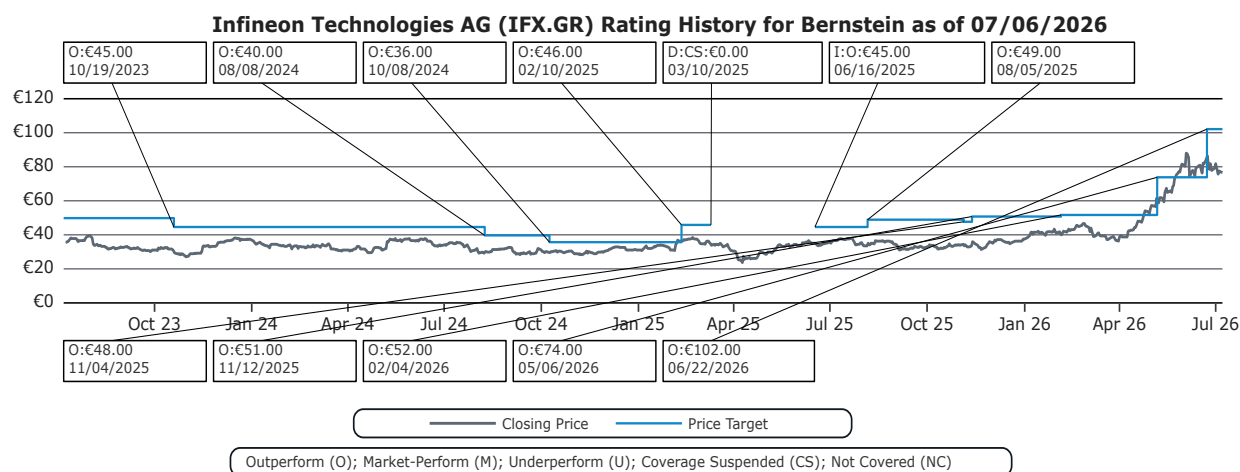
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